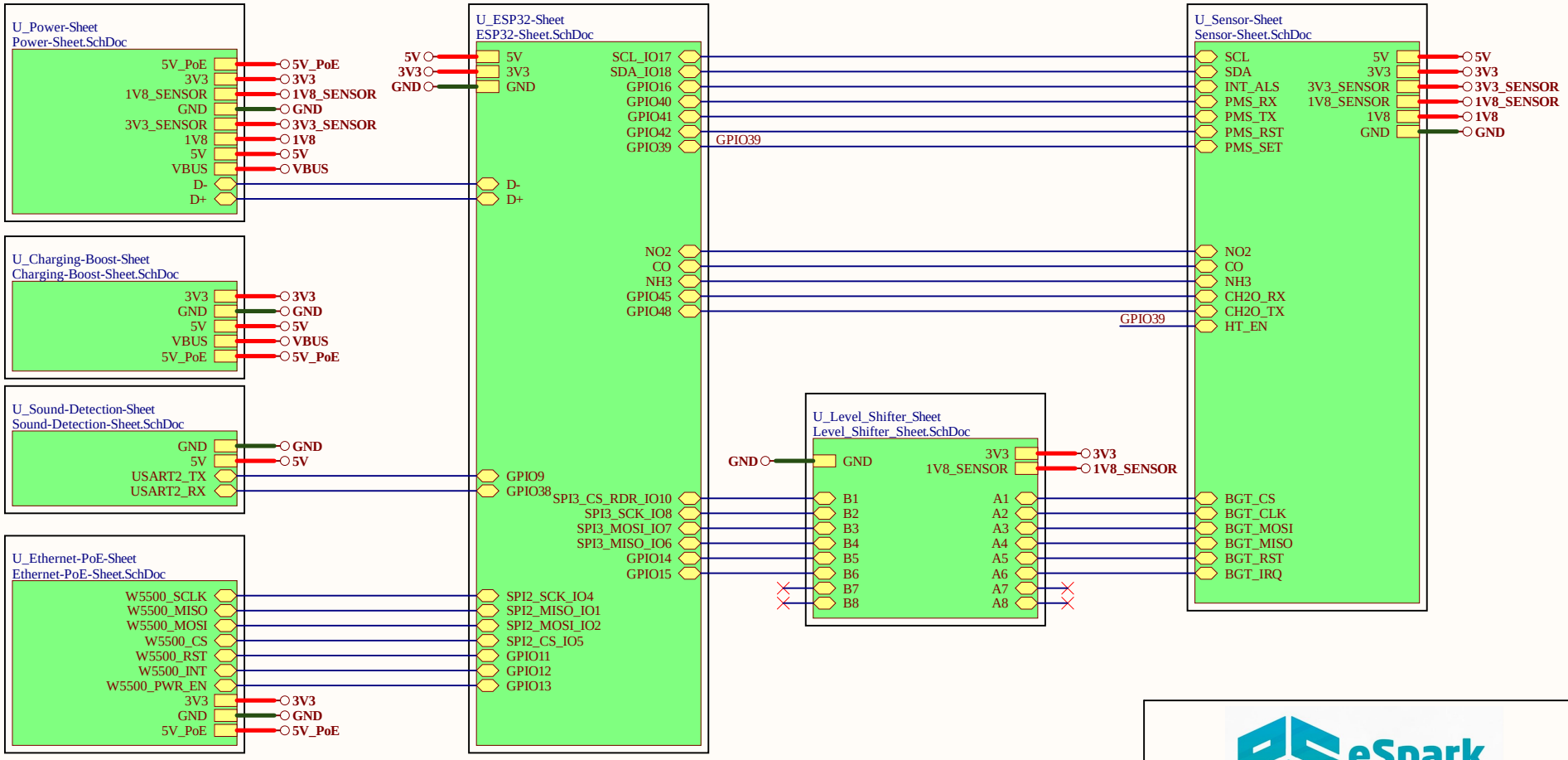
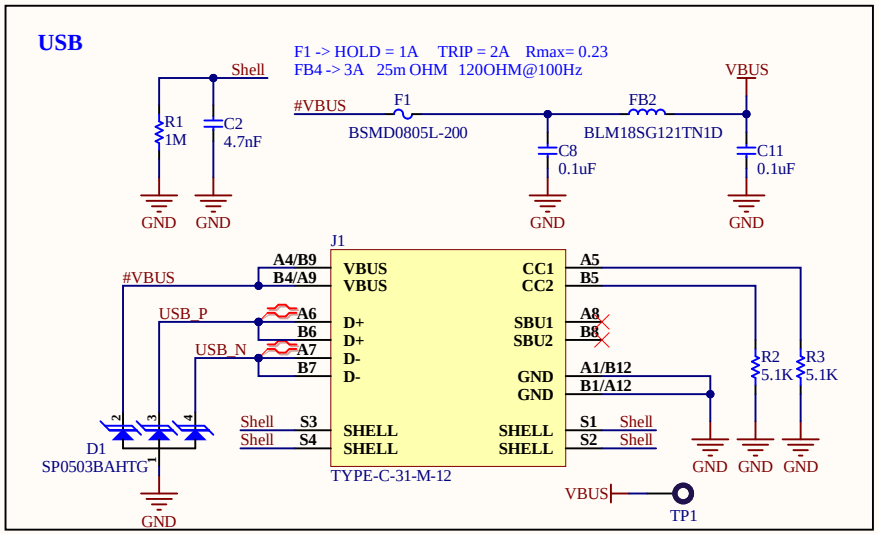
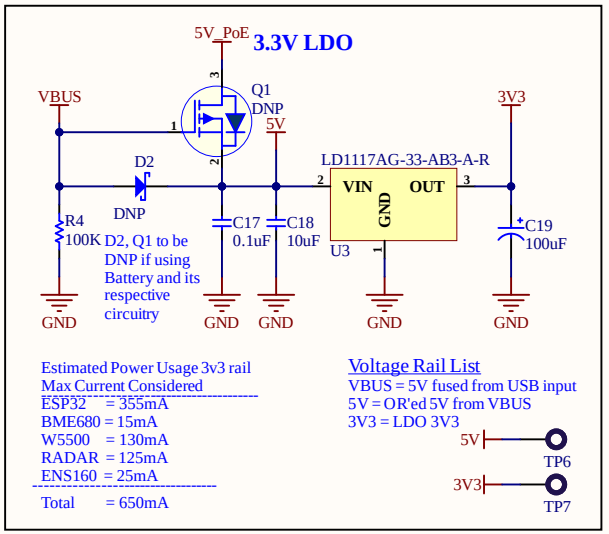
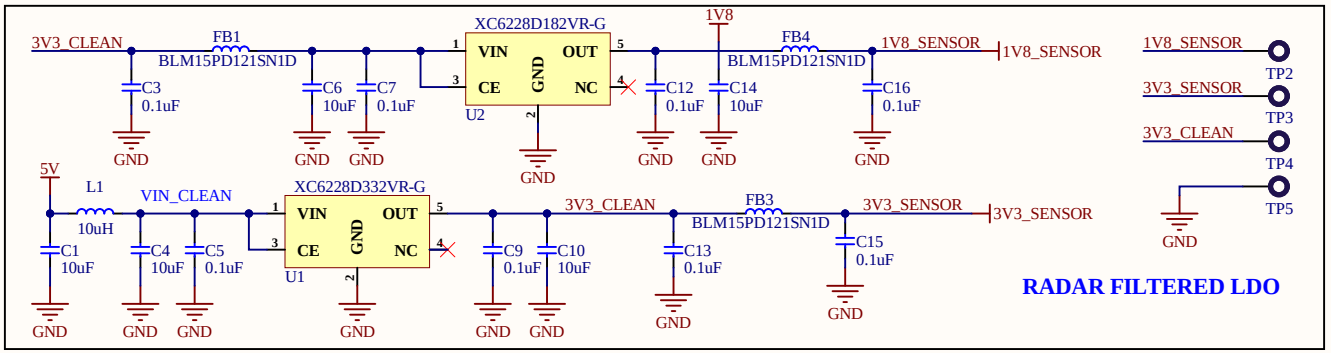


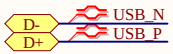
eSpark Consulting Group

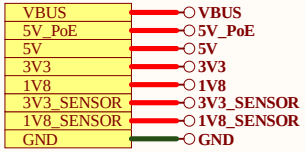
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Date: 14-Jun-23	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 8









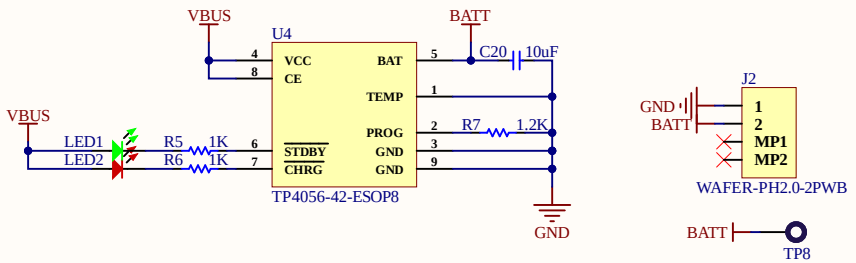


Power Section

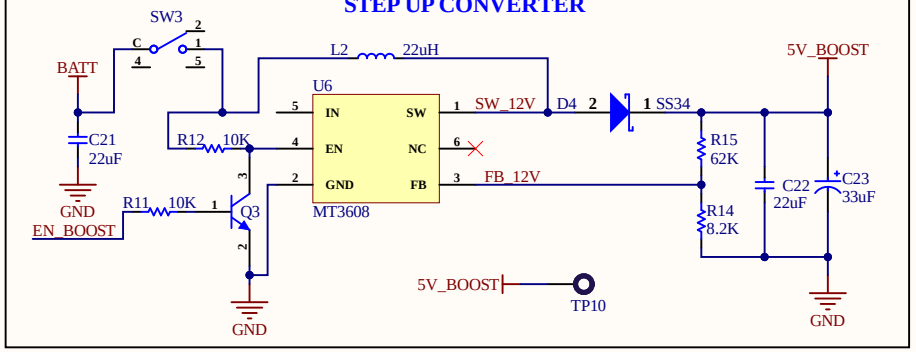
eSpark Consulting Group

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Date: 15-Jun-23	Page Num: 2
Drawn By: Huzaifa Ali	Page Total: 8

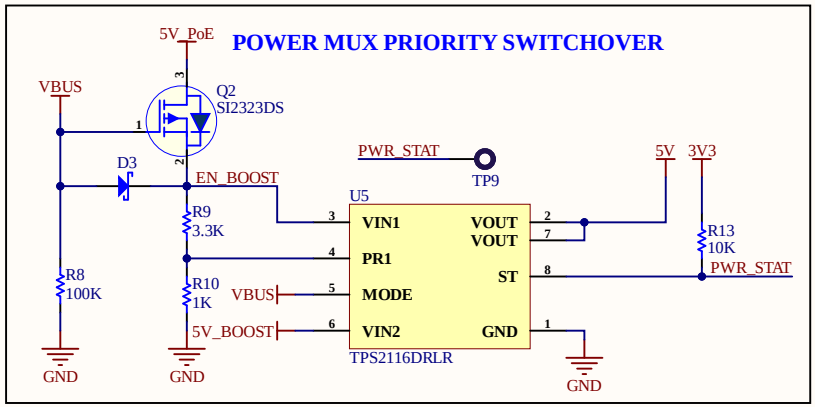
BATTERY CHARGING CIRCUIT

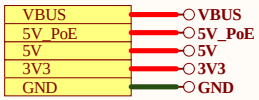


STEP UP CONVERTER



POWER MUX PRIORITY SWITCHOVER



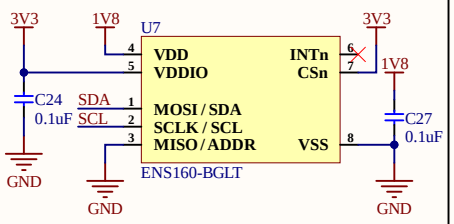


Charging Section

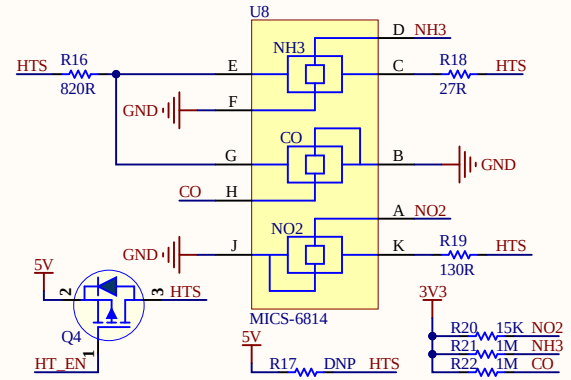
eSpark Consulting Group

Title: Charging-Boost-Sheet.SchDoc	
File: tracker-pro-airbnb-v1.0.PrjPcb	
Date: 15-Jun-23	Page Num: 3
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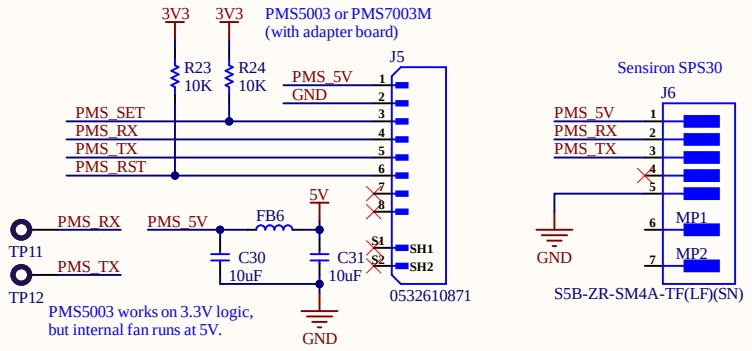
eCO2 TVOC AQI SENSOR



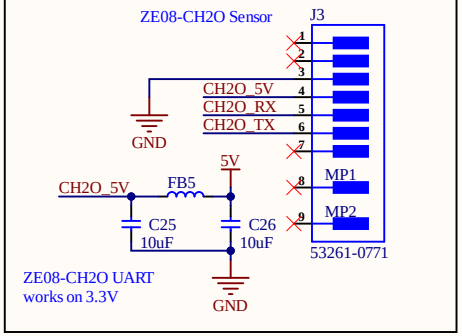
SUPPPORT SENSOR TO DETECT MARIJUANA



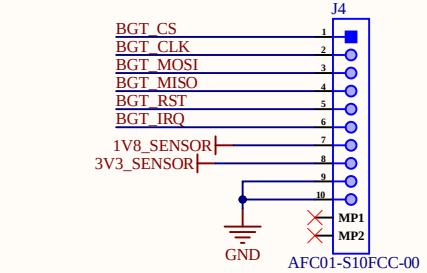
PM2.5 ENVIRONMENT SENSOR CIRCUIT



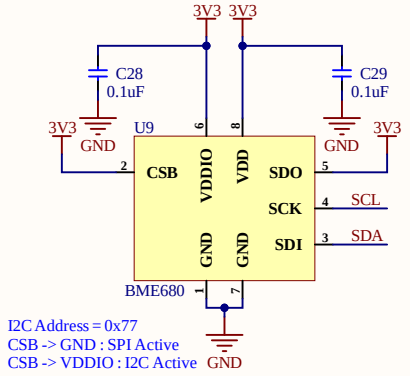
SUPPPORT SENSOR FOR MARIJUANA



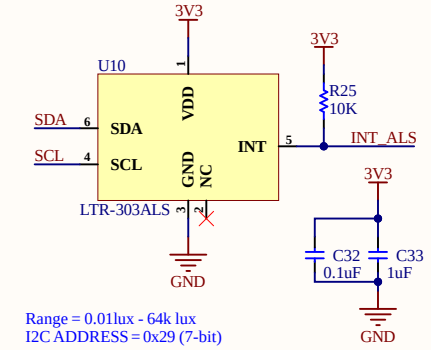
BGT60 RADAR SENSOR



TEMP HUM PRESSURE SENSOR



AMBIENT LIGHT SENSOR CIRCUIT



SDA
SCL
I2C pullups are given on Microcontroller-Sheet

BGT_CS
BGT_CLK
BGT_MOSI
BGT_MISO
BGT_RST
BGT_IRQ

PMS_SET
PMS_RX
PMS_TX
PMS_RST
INT_ALS

NO2
CO
NH3
HT_EN
CH2O_RX
CH2O_TX

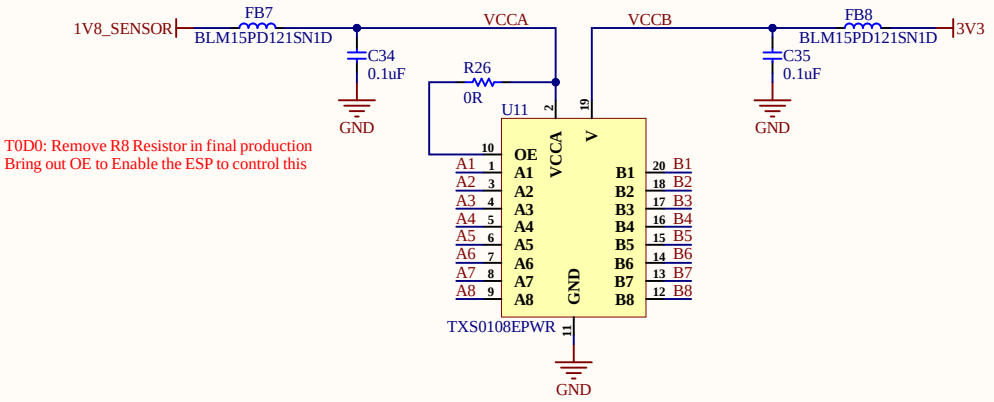
5V
3V3
1V8
3V3_SENSOR
1V8_SENSOR
GND

Sensors Section

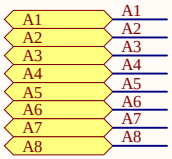
eSpark Consulting Group

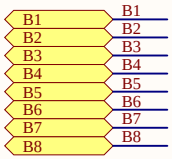
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File: tracker-pro-airbnb-v1.0.PrjPcb	
Date: 14-Jun-23	Page Num: 4
Drawn By: Huzaifa Ali	Page Total: 8

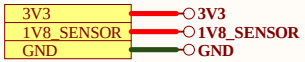
LEVEL SHIFTER



1.4V to 3.6V on A-PORT and 1.65V to 5.5V on B-PORT(VCCA < VCCB)



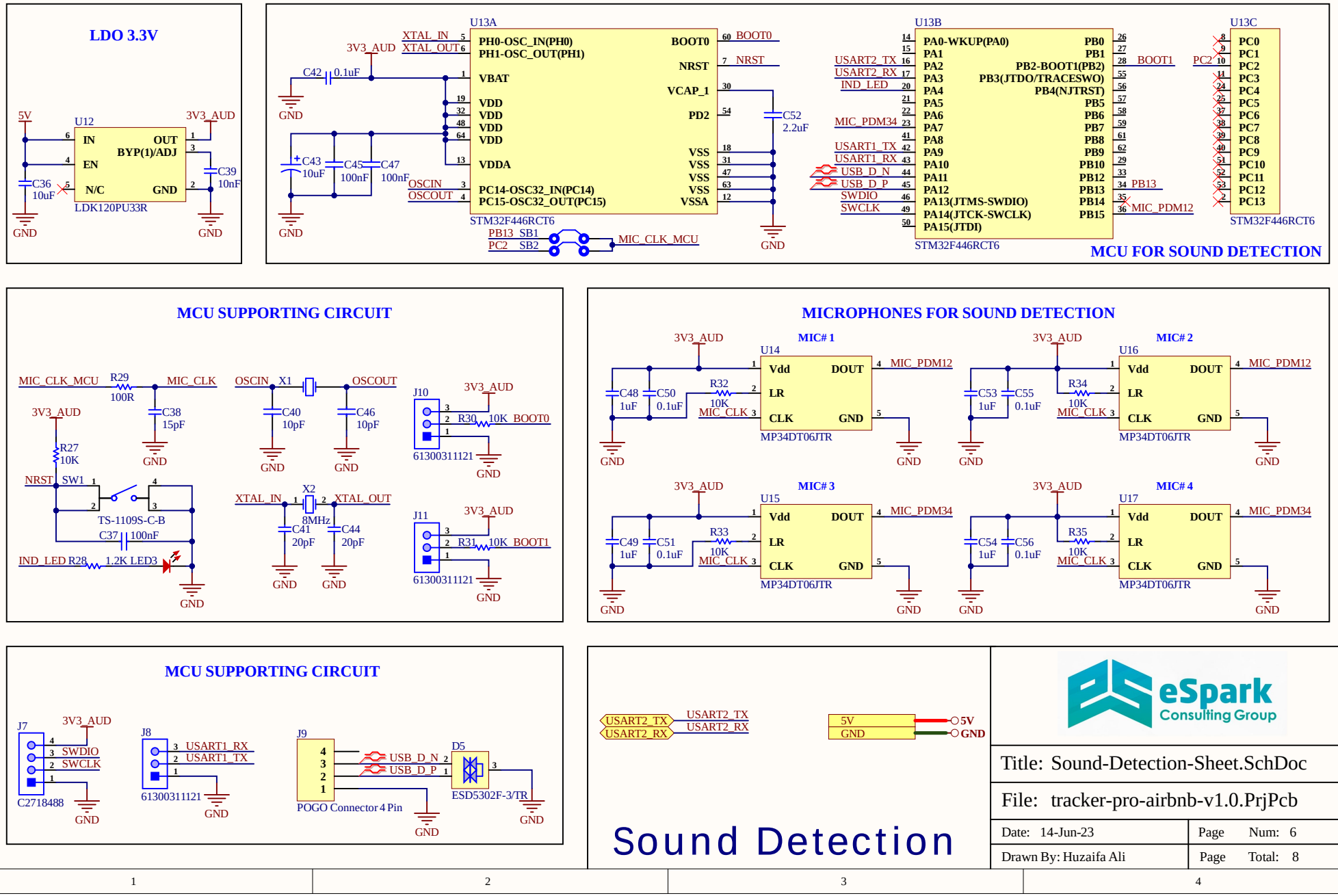


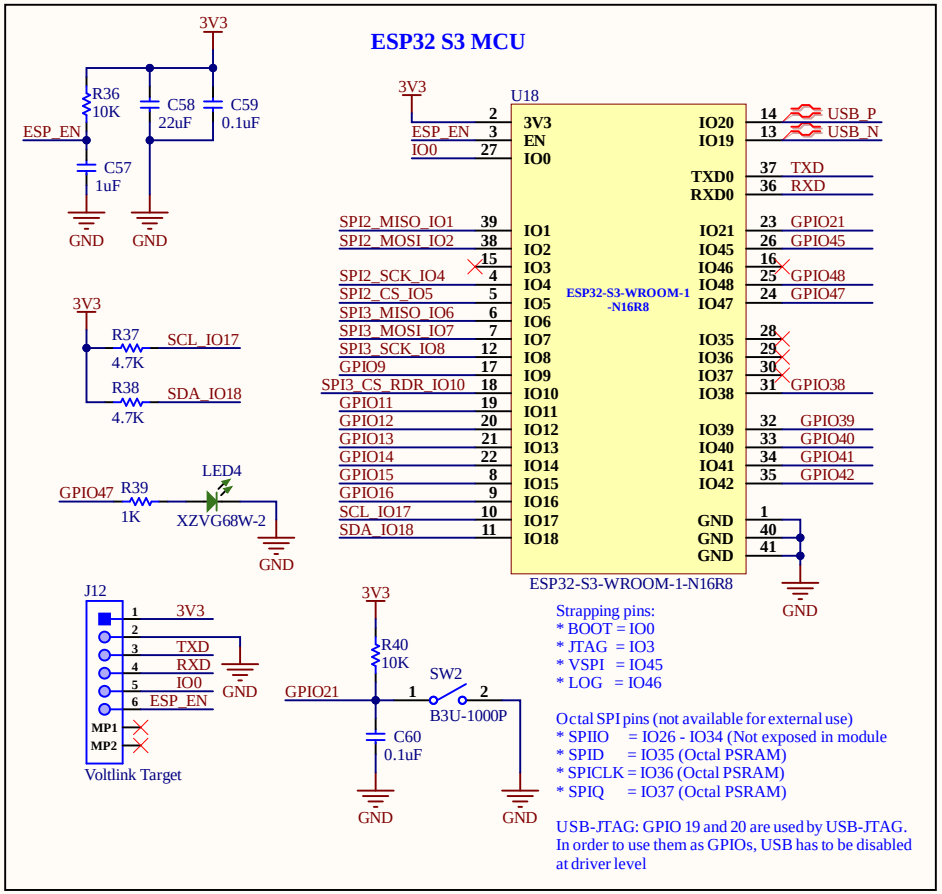


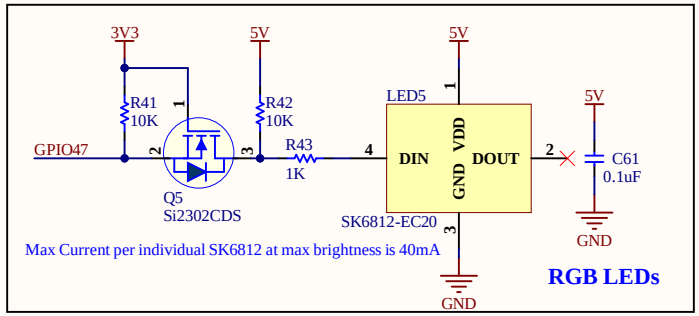
Level Shifter Section

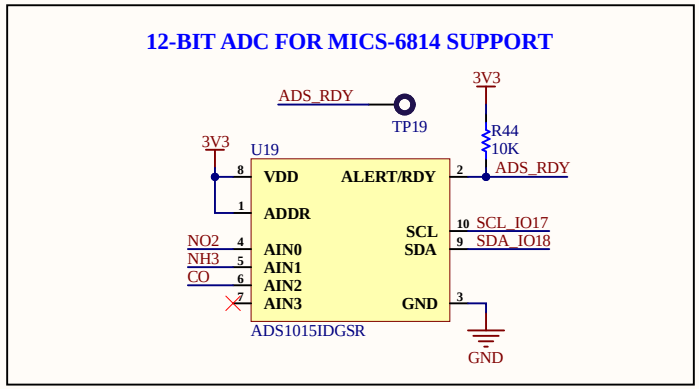
eSpark Consulting Group

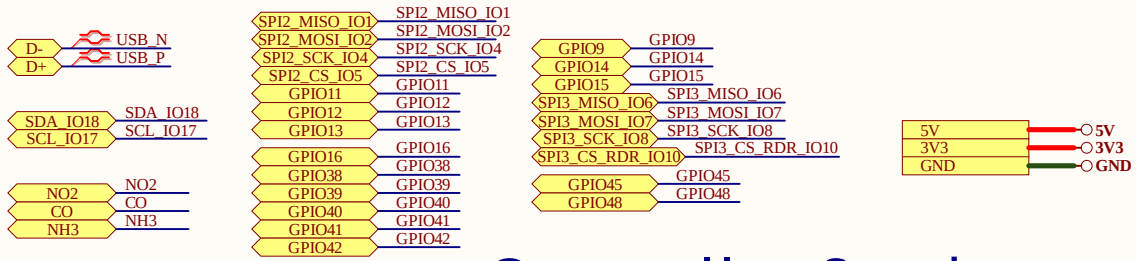
Title: Level_Shifter_Sheet.SchDoc	
File: tracker-pro-airbnb-v1.0.PrjPcb	
Date: 14-Jun-23	Page Num: 5
Drawn By: Huzaifa Ali	Page Total: 8


LDO 3.3V
5V
U12
IN
OUT
BYP(1)/ADJ
EN
N/C
GND
3V3_AUD
C36 10uF
C39 10nF
LDK120PU33R
GND
GND
U13A
XTAL_IN
XTAL_OUT
3V3_AUD
C42 0.1uF
PH0-OSC_IN(PH0)
PH1-OSC_OUT(PH1)
VBAT
VDD
VDD
VDD
VDD
VDDA
PC14-OSC32_IN(PC14)
PC15-OSC32_OUT(PC15)
BOOT0
NRST
VCAP_1
PD2
VSS
VSS
VSS
VSS
VSSA
C43 10uF
C45 100nF
C47 100nF
C52 2.2uF
OSCIN
OSCOUT
STM32F446RCT6
PB13 SB1
PC2 SB2
MIC_CLK_MCU
U13B
PA0-WKUP(PA0)
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13(JTMS-SWDIO)
PA14(JTCK-SWCLK)
PA15(JTDI)
PB0
PB1
PB2-BOOT1(PB2)
PB3(JTDO/TRACESWO)
PB4(NJTRST)
PB5
PB6
PB7
PB8
PB9
PB10
PB12
PB13
PB14
PB15
USART2_TX
USART2_RX
IND_LED
MIC_PDM34
USART1_TX
USART1_RX
USB_D_N
USB_D_P
SWDIO
SWCLK
BOOT1
PB13
MIC_PDM12
STM32F446RCT6
U13C
PC0
PC1
PC2
PC3
PC4
PC5
PC6
PC7
PC8
PC9
PC10
PC11
PC12
PC13
STM32F446RCT6
MCU FOR SOUND DETECTION
MCU SUPPORTING CIRCUIT
MIC_CLK_MCU
R29 100R
MIC_CLK
3V3_AUD
C38 15pF
R27 10K
GND
NRST
SW1
TS-1109S-C-B
C37 100nF
IND_LED
R28
1.2K
LED3
OSCIN
X1
OSCOUT
C40 10pF
C46 10pF
X2
XTAL_IN
XTAL_OUT
8MHz
C41 20pF
C44 20pF
J10
R30 10K
BOOT0
61300311121
J11
R31 10K
BOOT1
MICROPHONES FOR SOUND DETECTION
MIC# 1
MIC# 2
MIC# 3
MIC# 4
U14
U15
U16
U17
Vdd
DOUT
LR
CLK
GND
MP34DT06JTR
MIC_PDM12
MIC_PDM34
R32 10K
R33 10K
R34 10K
R35 10K
C48 1uF
C50 0.1uF
C49 1uF
C51 0.1uF
C53 1uF
C55 0.1uF
C54 1uF
C56 0.1uF
MCU SUPPORTING CIRCUIT
J7
SWDIO
SWCLK
C2718488
J8
USART1_RX
USART1_TX
61300311121
J9
USB_D_N
USB_D_P
POGO Connector 4 Pin
D5
ESD5302F-3/TR
USART2_TX
USART2_RX
5V
GND
Sound Detection
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Title: Sound-Detection-Sheet.SchDoc
File: tracker-pro-airbnb-v1.0.PrjPcb
Date: 14-Jun-23
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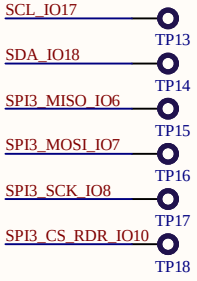






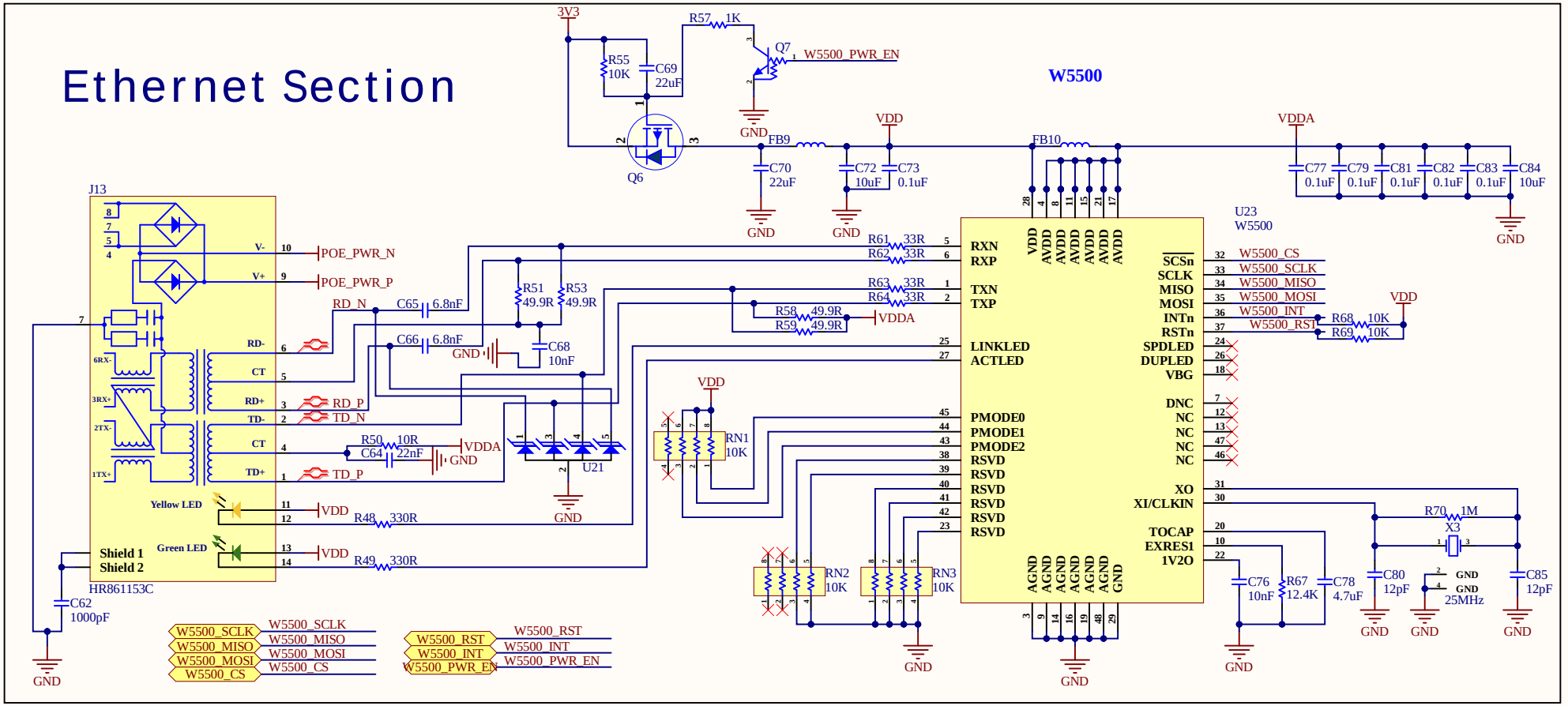


Controller Section



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Title: ESP32-Sheet.SchDoc	
File: tracker-pro-airbnb-v1.0.PrjPcb	
Date: 14-Jun-23	Page Num: 7
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Title:	Ethernet-PoE-Sheet.SchDoc
File:	tracker-pro-airbnb-v1.0.PrjPcb
Date: 15-Jun-23	Page Num: 8
Drawn By: Huzaifa Ali	Page Total: 8